

**UNIVERSITY
OF MALAYA**

**WQD7009
BIG DATA APPLICATIONS AND ANALYTICS**

SEM 1 2025/2026

GROUP ASSIGNMENT

**Title:
Exploring Obesity Trends Through Cloud-Based Analytics**

**LECTURER
TS. DR. RIYAZ AHAMED ARIYALURAN HABEEB MOHAMED**

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Group 21 (OCC 2)

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1.0. Introduction

The global obesity problem is becoming severe, and it is crucial to understand the behavioral and lifestyle factors behind it. The development of big data and cloud computing technologies has provided new opportunities for in-depth analysis of health data. This project aims to build a cloud-based data analysis framework to explore the specific impact of dietary and exercise habits on obesity levels.

1.1. Objectives

The core objectives of this project:

1. Build a full-cycle framework for data analysis based on Google Cloud Platform (GCP), achieving complete lifecycle management from data collection, storage, processing to analysis and visualization.
2. Complete data preprocessing and feature engineering in BigQuery to form standardized analysis tables, providing a data foundation for subsequent processes.
3. Use BigQuery and Colab for predictive analysis, exploring key behavioral factors influencing obesity levels and predictive models.
4. Build an interactive visualization dashboard to enable multi-dimensional data exploration and descriptive analysis, revealing relationships among features.
5. Evaluate the framework's performance through metrics such as query latency, throughput, and data consistency, verifying the practicality and efficiency of the cloud platform in data engineering.

1.2. Dataset Description

The dataset of this project is the "Obesity Level Assessment Based on Dietary Habits and Physical Condition" from the UCI database, containing 2,111 records and 17 attributes. It mainly covers four types of information:

1. **Demographic and Physical Indicators:** gender, age, height, weight.
2. **Dietary Behaviors:** frequency of intake of high-calorie foods, frequency of vegetable consumption.
3. **Exercise and Lifestyle Habits:** frequency of physical activity, screen time, frequency of alcohol consumption.
4. **Target Variables:** Seven types of obesity levels (from underweight to grade 3 obesity).

This dataset has clear structure with few missing values. It can support full life cycle data analysis and behavioral pattern exploration.

2.0. Meeting Minutes Report

2.1 Meeting Minute #1

Date: 30 November 2025

Time: 12.58 PM

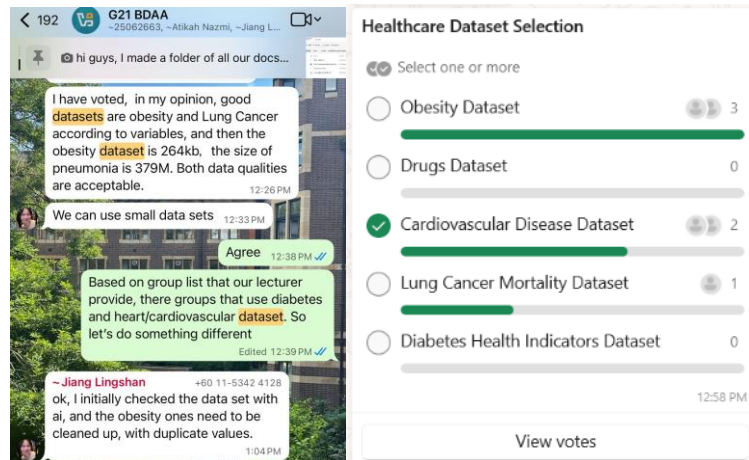
Location: WhatsApp Group Discussion

Attendee(s): All present

Main Agenda(s): Dataset evaluation and selection and Project planning

Proceedings Summary:

1. Dataset Proposal Phase (23-30 Nov 2025)
 - Each member proposed one healthcare dataset via shared Google Sheet.
 - <https://docs.google.com/spreadsheets/d/1uYQGF3EOouNcDuqJSihTPu-uClprkholj5F-PliVYJk/edit?gid=1595732739#gid=1595732739>
 - Some key considerations are stated for each dataset such as the dataset's relevance to the project objective, data quality and completeness, size and structure, variety and richness, applicability to the healthcare domain, potential for meaningful data exploration, compatibility with analytical tools, opportunities for innovation, and its interoperability or integration potential.
2. Voting Results (30 Nov 2025 via WhatsApp poll)



- The majority of group members selected the “Obesity Dataset” based on relevant considerations.

3. Task Distribution

Ling Long				Ling Shan		
intro	5			Comparison (on premises/cloud-based tools) & Justification	10	
data visualization (development & reporting)	5			data architecture diagram (drafting)	5	
evaluation metric with graph (based on Visualization)	3.25					
Total Marks	13.25			Total Marks	15	
Atikah				Wei Shyong		
meeting minute 1	2.5			data architecture diagram (confirmation)	7.5	
data preparation (reporting)	2.5			meeting minute 2	2.5	
data analytics (development)(Predictive analysis)	2.5			data analytics (reporting) (Predictive analysis)	2.5	
evaluation metric with graph (based on Analytica)	5.75			data storage (reporting)	2.5	
Total Marks	13.25			Total Marks	15	
Madina						
data ingestion (development & reporting)	5					
data preparation (development)	2.5					
data storage (development)	2.5					
data architecture (idea generation)	2.5					
evaluation metric with graph	1					
Total Marks	13.5					

- The task distribution is aligned with the allocated marks; however, the final marks awarded are solely based on the lecturer's evaluation.

Action Item: Start data architecture diagram using LucidChart

Prepared by: Atikah binti Ahmad Nazmi

Date: 3 December 2025

2.2 Meeting Minute #2

Date: 17 December 2025

Time: 4.30PM

Location: Microsoft Teams

Attendee(s): All present

Absentee(s): -

Main Agenda(s): Go through the Data Architecture and Finalize the Report

Proceedings Summary:

1. Explained the data architecture process from uploading the data to cloud until standardizing the data.
2. Highlighted the need to reduce word counts in the report and proceed with task distribution.
3. Proposed to add in Conclusion and Reference sections.
4. Final review set on Friday noon.

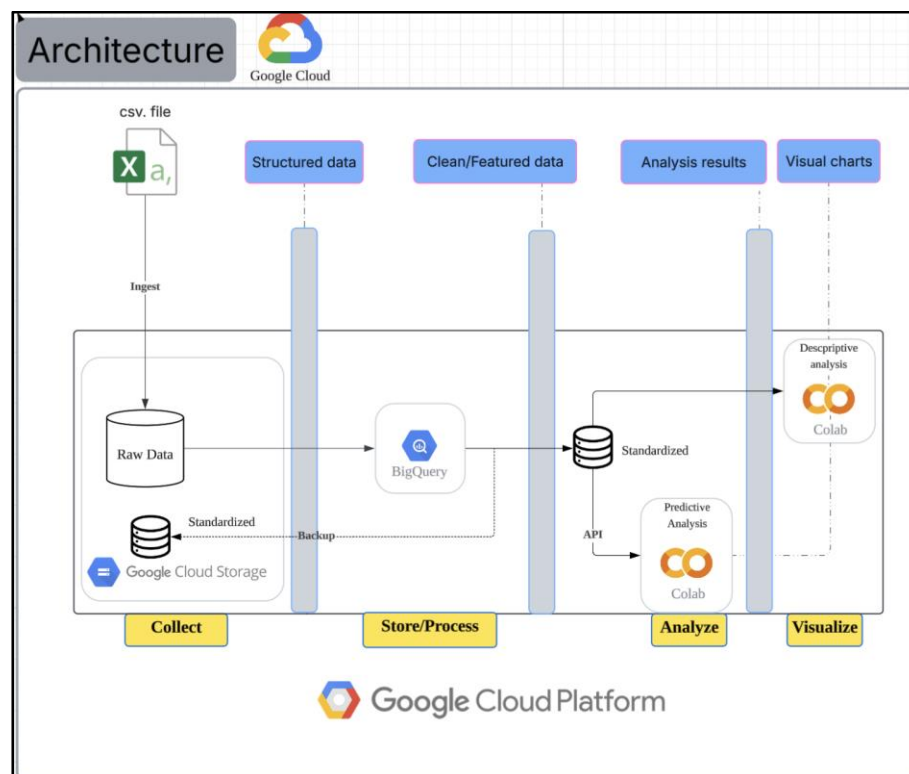
Action Item:

- Condense content to reduce the word counts – All members
- Complete the Conclusion, Reference, Meeting Minute sections – Wei Shyong
- Teams recording: https://365umedumy-my.sharepoint.com/:v/g/personal/24235116_siswa365_um_edu_my/IQAeXUUbZ3rDS07pVoXFloEfAQGKIProL9EElikDHCUDPVs

3.0. Data Architecture

GCP-based Architecture Explanation

We break the typical Data Life Cycle (DLC) into 4 main stages: Collect, Store and Process, Analyze, Active. In the Collect stage, we ingest our input file (in CSV format) into Google Cloud Storage (GCS). The raw data is stored in its original format but in the cloud environment and GCS acts as the Data Lake. Next, we use BigQuery to perform ETL (Extract, Transform, Load), where the data is read from GCS for transformation. BigQuery also writes and stores the clean data as the Standardized Data. The data has now become the source for analysis, and we can use API to link the data to Colab for further analysis. As for the Active stage, we use Colab again to visualize our insights via charts and graphs.



4.0. Tools Selection Justification

The table below outlines the differences in DLC stages between Traditional Methods and GCP Framework.

Stage	Key Objective	Traditional Methods	GCP Framework
Collect	Orderliness and efficiency in ingestion.	Manual work leads to version confusion, file loss, and lack of traceability.	Centralized cloud storage (GCS) ensures traceability through logs and avoids version conflicts.
Store	Low-cost, safe, stable, and easy to query.	MySQL requires manual installation, regular backups, and is prone to data loss from failures.	BigQuery offers a serverless architecture with automatic backups, high query efficiency, and no maintenance overhead.
Process	Converting raw data into standardized analytical forms.	Apache Spark clusters require heavy deployment, maintenance, and complex optimization.	BigQuery provides fast starts, built-in monitoring, and reduces operation and maintenance work.
Analyze	Extracting patterns and insights via statistical methods.	Local Pandas has memory limitations, weak machine learning (ML) capabilities, and high collaboration thresholds.	Colab offers a collaborative environment, low ML thresholds, and real-time result sharing with deep GCP integration.
Visualize	Developing interactive dashboards for actionable insights.	Static Power BI exports are not dynamic and often face version control issues.	Colab integrated with Plotly supports interactive, real-time filtering mapped directly to BigQuery for data consistency.

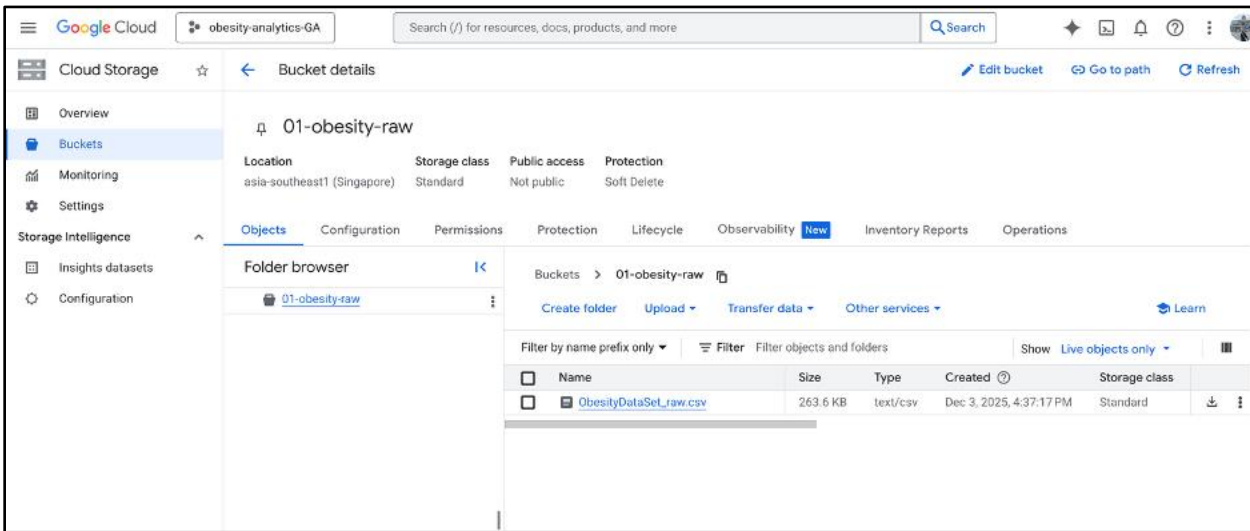
5.0. Implementation of the Proposed Framework

The implementation in Google Cloud Platform (GCP) started by creating a Project ID, which acts as a unique identifier to differentiate between projects. It serves as a container that groups all cloud resources such as storage buckets, BigQuery datasets, API, and access permissions. The project ID is set to be “obesity-analytics-GA”.

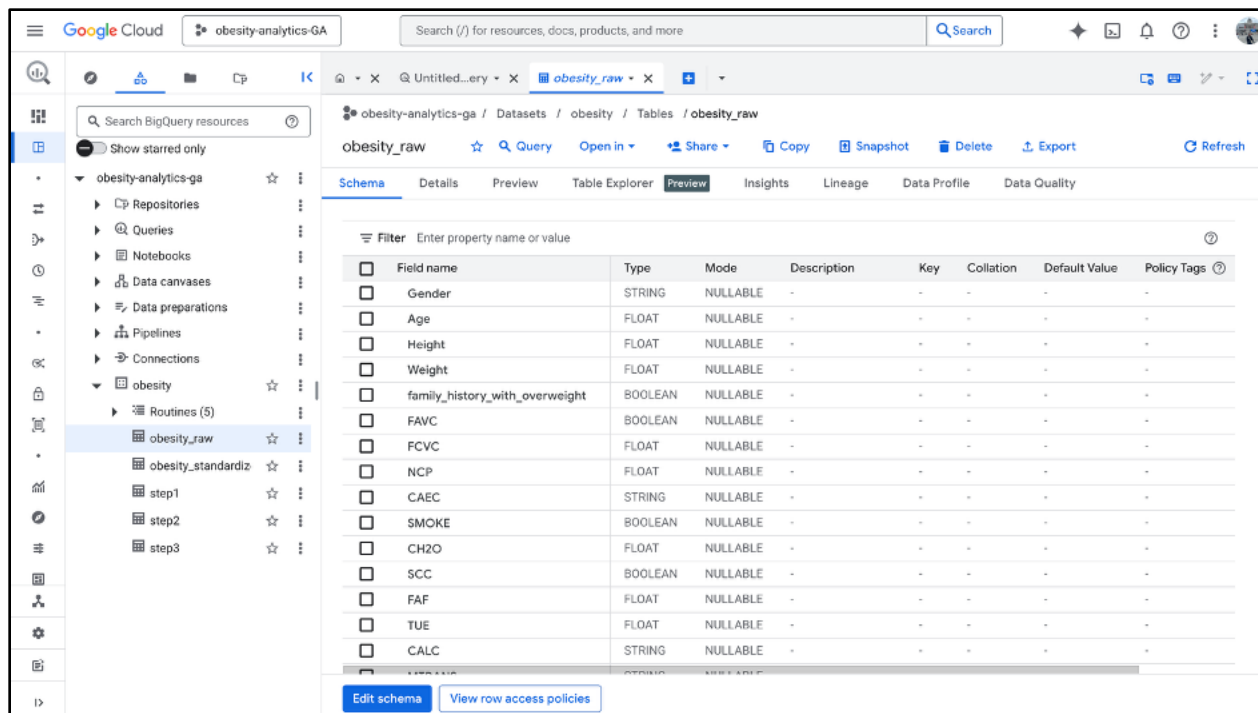
5.1. Data Ingestion

The raw obesity dataset is provided in CSV format, and it acts as an initial input to our GCP. Since the dataset is not real-time data, we upload it from the local environment into Google Cloud Storage (GCS) and consider it a raw layer of data. The purpose of uploading this data into GCS is to ensure a seamless integration with BigQuery.

The ingestion process started by creating a bucket inside GCS, and we named it “01-obesity-raw”. This bucket is dedicated to store only raw data and act as the first layer of the overall data warehouse. We set the geographic region as “asia-southeast1 (Singapore)” and this region supposed to be set uniformly set with BigQuery to reduce data transfer latency. Moreover, public access prevention enables us to make sure the data is not exposed to the public internet. After setting up the bucket, the data can be uploaded safely into the bucket as shown in figure below.

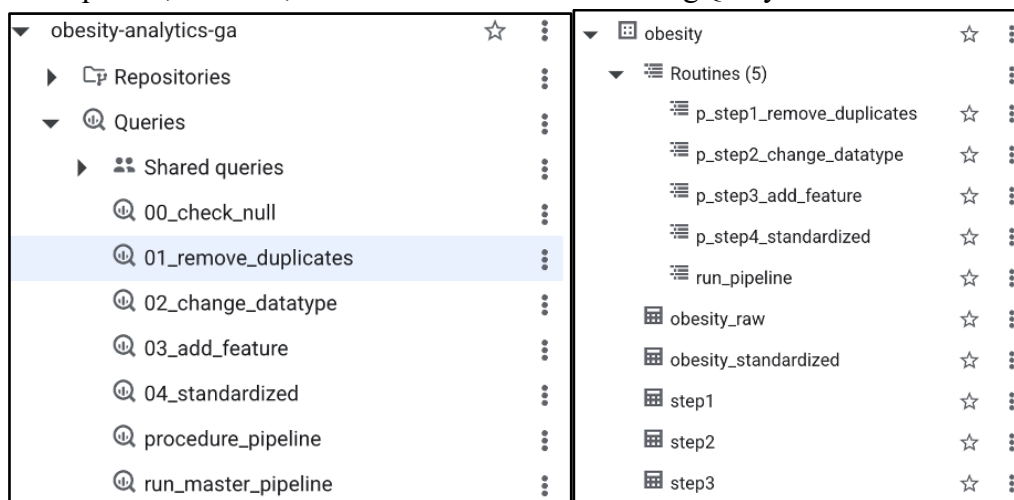


Next, raw data from GCS can now be loaded into Google BigQuery as a raw table named “obesity_raw”. During the loading process, no data transformation or cleaning has been done yet thus, it preserved the original structure and content of the dataset. We can refer to the original data whenever needed. Figure below shows the ingested table inside BigQuery.



5.2. Data Pre-processing

Data preprocessing is implemented using **BigQuery stored procedures** to ensure scalable and dynamic transformation workflow. The preprocessing stage aims to transform raw data from “**obesity_raw**” table into a standardized form using a multi-step SQL pipeline. Each processing task is encapsulated within a dedicated stored procedure for ease of maintenance. Before saving the cleaned data into “**obesity_standardized**” table, there will be some intermediate output tables (step1, step2, step3). This process ensures data lineage and traceability. Furthermore, a master procedure “**run_pipeline**” will orchestrate the execution of all processing steps. Below pictures show a list of queries, routines, and tables saved inside our BigQuery environment.



Before performing data cleaning, initial data quality checks must be performed, such as checking the datatype and null count. This step provides an early detection of data quality issues without modifying the original data. Figure below shows the datatype of the original table and the count of nulls.

obesity-analytics-ga / Datasets / obesity / Tables / obesity_raw

☆ obesity_raw 🔍 Query ⌵ Open in 👤 Share 📄 Copy 📄 Sna

Schema Details Preview Table Explorer Preview Insights Lineage

Filter Enter property name or value

☐	Field name	Type	Mode	Description
☐	Gender	STRING	NULLABLE	-
☐	Age	FLOAT	NULLABLE	-
☐	Height	FLOAT	NULLABLE	-
☐	Weight	FLOAT	NULLABLE	-
☐	family_history_with_overweight	BOOLEAN	NULLABLE	-
☐	FAVC	BOOLEAN	NULLABLE	-
☐	FCVC	FLOAT	NULLABLE	-
☐	NCP	FLOAT	NULLABLE	-
☐	CAEC	STRING	NULLABLE	-
☐	SMOKE	BOOLEAN	NULLABLE	-
☐	CH2O	FLOAT	NULLABLE	-
☐	SCC	BOOLEAN	NULLABLE	-
☐	FAF	FLOAT	NULLABLE	-
☐	TUE	FLOAT	NULLABLE	-
☐	CALC	STRING	NULLABLE	-
☐	MTRANS	STRING	NULLABLE	-
☐	NObesyesdad	STRING	NULLABLE	-

obesity_raw X 00_che... ull X +

🔍 00_check_null ⏮ Run 📄 Save query ⬇ Download 👤 Share ⌚ Schedule ⌵ Open in ⚙ More

1 SELECT
2 COUNTIF(Gender IS NULL) AS null_gender,
3 COUNTIF(Age IS NULL) AS null_age,
4 COUNTIF(Height IS NULL) AS null_height,
5 COUNTIF(Weight IS NULL) AS null_weight,
6 COUNTIF(family_history_with_overweight IS NULL) AS null_history,
7 COUNTIF(FAVC IS NULL) AS null_favc,
8 COUNTIF(FCVC IS NULL) AS null_fcvc,
9 COUNTIF(NCP IS NULL) AS null_ncp,
10 COUNTIF(CH2O IS NULL) AS null_ch2o,
11 COUNTIF(FAF IS NULL) AS null_faf,
12 COUNTIF(TUE IS NULL) AS null_tue,
13 COUNTIF(CAEC IS NULL) AS null_caec,
14 COUNTIF(SMOKE IS NULL) AS null_smoke,
15 COUNTIF(SCC IS NULL) AS null_scc,
16 COUNTIF(CALC IS NULL) AS null_calc,

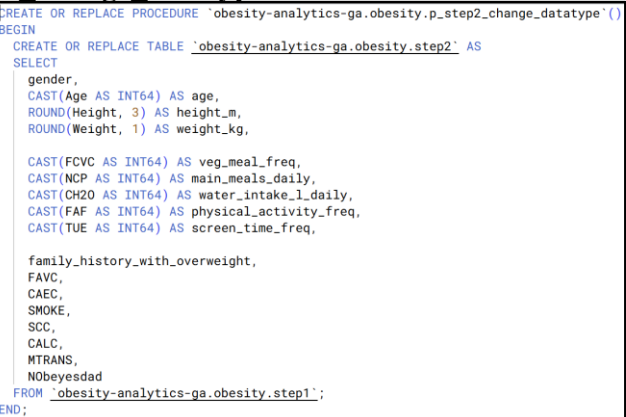
✔ Query completed

Query results Save results Open in

Job information Results Visualization JSON Execution details Execution graph

null_gender	null_age	null_height	null_weight	null_history	null_favc	null_fcvc	null_ncp	null_c
0	0	0	0	0	0	0	0	

As we can see, there is no null value in each column, thus the data is almost cleaned. Now, we will perform data preprocessing by dividing queries into steps of cleaning procedures. The table below shows the steps and its description.

Data Cleaning Procedure	Description
01_remove_duplicates  <pre> 1 CREATE OR REPLACE PROCEDURE `obesity-analytics-ga.obesity.p_step1_remove_duplicates`() 2 BEGIN 3 CREATE OR REPLACE TABLE `obesity-analytics-ga.obesity_step1` AS 4 SELECT DISTINCT * 5 FROM `obesity-analytics-ga.obesity_raw`; 6 END; </pre>	Duplicate rows are removed using SELECT DISTINCT from raw dataset. The deduplicated output will be stored in the intermediate table named “step1”. Initially, there are a total of 2,111 rows and since there are 24 duplicated rows, the number of rows reduced to 2,087. Finally, this query was saved into a stored procedure named “p_step1_remove_duplicates”.
02_change_datatype  <pre> 1 CREATE OR REPLACE PROCEDURE `obesity-analytics-ga.obesity.p_step2_change_datatype`() 2 BEGIN 3 CREATE OR REPLACE TABLE `obesity-analytics-ga.obesity_step2` AS 4 SELECT 5 gender, 6 CAST(Age AS INT64) AS age, 7 ROUND(Height, 3) AS height_m, 8 ROUND(Weight, 1) AS weight_kg, 9 10 CAST(FCVC AS INT64) AS veg_meal_freq, 11 CAST(NCP AS INT64) AS main_meals_daily, 12 CAST(CH20 AS INT64) AS water_intake_l_daily, 13 CAST(FAF AS INT64) AS physical_activity_freq, 14 CAST(TUE AS INT64) AS screen_time_freq, 15 16 family_history_with_overweight, 17 FAVC, 18 CAEC, 19 SMOKE, 20 SCC, 21 CALC, 22 MTRANS, 23 NObesydad 24 FROM `obesity-analytics-ga.obesity_step1`; 25 END; </pre>	Most of the columns are cast from FLOAT to INTEGER. The output is stored in an intermediate table named “step2”. This query is saved into a stored procedure named “p_step2_change_datatype”.
03_add_feature  <pre> 1 CREATE OR REPLACE PROCEDURE `obesity-analytics-ga.obesity.p_step3_add_feature`() 2 BEGIN 3 CREATE OR REPLACE TABLE `obesity-analytics-ga.obesity_step3` AS 4 SELECT 5 *, 6 ROUND(weight_kg / (height_m * height_m), 2) AS bmi 7 FROM `obesity-analytics-ga.obesity_step2`; 8 END; </pre>	Based on column height and weight, Body Mass Index (BMI) is derived. Featured-enriched data is stored in an intermediate table named “step3” and stored in a procedure named “p_step3_add_feature”.
04_standardized	The naming conventions of columns were taken place to help data analysts understand the schema better. The final output is stored inside “obesity_standardized” table, and the query is saved into “p_step4_standardized”.

```

CREATE OR REPLACE PROCEDURE `obesity-analytics-ga.obesity.p_step4_standardized`()
BEGIN
  CREATE OR REPLACE TABLE `obesity-analytics-ga.obesity.obesity_standardized` AS
  SELECT
    gender,
    age,
    height_m,
    weight_kg,
    bmi,

    veg_meal_freq,
    main_meals_daily,
    water_intake_l_daily,
    physical_activity_freq,
    screen_time_freq,

    family_history_with_overweight,
    FAVC AS high_caloric_food_intake,
    SMOKE AS is_smoker,
    SCC AS monitors_calories,

    CALC AS alcohol_intake,
    CAEC AS eating_between_meals,
    MTRANS AS transport_mode,
    NObeyesdad AS obesity_category
  FROM `obesity-analytics-ga.obesity.step3`;
END;

```

Pipeline automation and execution

All processing steps are compiled into a master procedure named “**run_pipeline**”. The pipeline consists of 5 stored procedures, and a dedicated query named “**run_master_pipeline**” will trigger the execution of the entire pipeline. The pipeline can be re-run on demand with a single command or through query scheduling. Figure below shows the compilation of procedures in “**procedure_pipeline**” query and the results of executing “**run_master_pipeline**” query.

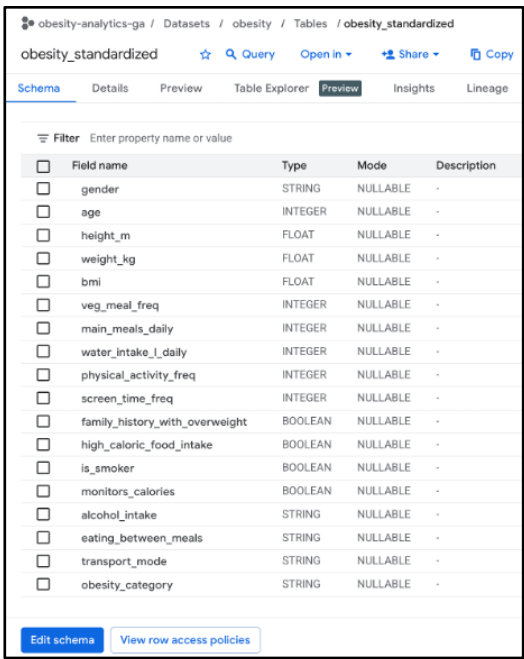
```

procedure_pipeline
1 CREATE OR REPLACE PROCEDURE `obesity-analytics-ga.obesity.run_pipeline`()
2 BEGIN
3   CALL `obesity-analytics-ga.obesity.p_step1_remove_duplicates`();
4   CALL `obesity-analytics-ga.obesity.p_step2_change_datatype`();
5   CALL `obesity-analytics-ga.obesity.p_step3_add_feature`();
6   CALL `obesity-analytics-ga.obesity.p_step4_standardized`();
7 END;

```

run_master_pipeline						
1 CALL `obesity-analytics-ga.obesity.run_pipeline`();						
Query completed						
Using on-demand processing quota						
All results						
Elapsed time		Statements processed		Job status		
15 sec		4		SUCCESS		
Status	End time	SQL	Stages completed	Bytes pr	Action	
✓	3:44 PM Procedure	CREATE OR REPLACE TABLE `obesity-analytics-ga.obesity.step1` AS	4	274.37 K	View results	
✓	3:44 PM Procedure	CREATE OR REPLACE TABLE `obesity-analytics-ga.obesity.step2` AS	3	271.25 K	View results	
✓	3:44 PM Procedure	CREATE OR REPLACE TABLE `obesity-analytics-ga.obesity.step3` AS	3	271.25 K	View results	
✓	3:44 PM Procedure	CREATE OR REPLACE TABLE `obesity-analytics-ga.obesity.obesity_standardized` AS	3	287.56 K	View results	

Finally, a cleaned and processed table is stored inside “**obesity_standardized**” as shown in figure below.



The screenshot shows the BigQuery Schema page for the table 'obesity_standardized'. The table has 20 columns with various data types and NULLABLE modes. The columns are: gender (STRING), age (INTEGER), height_m (FLOAT), weight_kg (FLOAT), bmi (FLOAT), veg_meal_freq (INTEGER), main_meals_daily (INTEGER), water_intake_l_daily (INTEGER), physical_activity_freq (INTEGER), screen_time_freq (INTEGER), family_history_with_overweight (BOOLEAN), high_caloric_food_intake (BOOLEAN), is_smoker (BOOLEAN), monitors_calories (BOOLEAN), alcohol_intake (STRING), eating_between_meals (STRING), transport_mode (STRING), and obesity_category (STRING). All columns are NULLABLE.

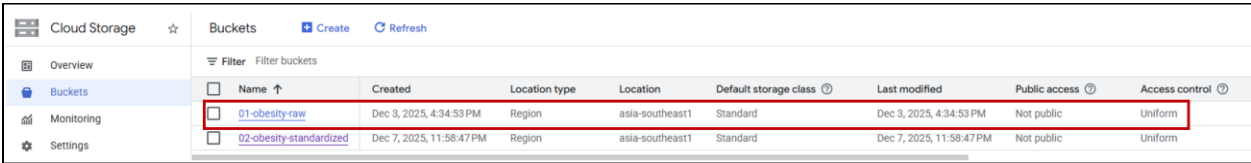
Field name	Type	Mode	Description
gender	STRING	NULLABLE	-
age	INTEGER	NULLABLE	-
height_m	FLOAT	NULLABLE	-
weight_kg	FLOAT	NULLABLE	-
bmi	FLOAT	NULLABLE	-
veg_meal_freq	INTEGER	NULLABLE	-
main_meals_daily	INTEGER	NULLABLE	-
water_intake_l_daily	INTEGER	NULLABLE	-
physical_activity_freq	INTEGER	NULLABLE	-
screen_time_freq	INTEGER	NULLABLE	-
family_history_with_overweight	BOOLEAN	NULLABLE	-
high_caloric_food_intake	BOOLEAN	NULLABLE	-
is_smoker	BOOLEAN	NULLABLE	-
monitors_calories	BOOLEAN	NULLABLE	-
alcohol_intake	STRING	NULLABLE	-
eating_between_meals	STRING	NULLABLE	-
transport_mode	STRING	NULLABLE	-
obesity_category	STRING	NULLABLE	-

5.3. Data Storage

During the Store/Process stage, the architecture utilizes a layered storage approach within BigQuery to maintain data lineage. Data flows through intermediate tables where preprocessing steps are performed. This process produces a "standardized" table containing the analytics-ready dataset. For versioning and external sharing, this output is exported back to a separate GCS bucket.

1. Raw file storage layer (GCS)

- **Raw bucket**
 - Stores the original obesity dataset **in CSV format**.
 - Purpose:
 - Acts as the **data lake landing zone**.
 - Preserves the **original file** for auditability and reproducibility.
 - Enable reloading if the BigQuery table needs to be rebuilt.

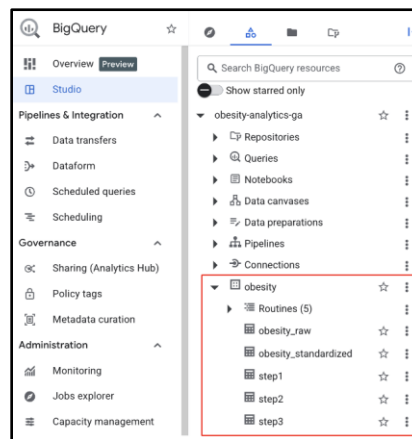


The screenshot shows the Google Cloud Storage Buckets page. It lists two buckets: '01-obesity-raw' and '02-obesity-standardized'. Both buckets are located in 'asia-southeast1' and have a 'Standard' storage class. The '01-obesity-raw' bucket was created on Dec 3, 2025, and the '02-obesity-standardized' bucket was created on Dec 7, 2025. Both are 'Not public' and have 'Uniform' access control.

Name	Created	Location type	Location	Default storage class	Last modified	Public access	Access control
01-obesity-raw	Dec 3, 2025, 4:34:53 PM	Region	asia-southeast1	Standard	Dec 3, 2025, 4:34:53 PM	Not public	Uniform
02-obesity-standardized	Dec 7, 2025, 11:58:47 PM	Region	asia-southeast1	Standard	Dec 7, 2025, 11:58:47 PM	Not public	Uniform

2. Structured storage & processing layer (BigQuery)

- **Raw Table**
 - **Table:** obesity_raw
 - Loaded from the raw GCS bucket into BigQuery **without transformation**.
 - Purpose:
 - Serves as the **source-of-truth**.
 - 2,111 rows and 17 columns.
- **Intermediate Tables**
 - **Tables:** step1, step2, step3 (based on stored procedure outputs)
 - Purposes:
 - Store results of each preprocessing step separately.
 - Improve **traceability**.
 - Support **separation of concerns** and validation.
- **Standardized Table**
 - **Table:** obesity_standardized
 - Purposes:
 - Contains the final cleaned and standardized dataset with consistent schema and engineered features.
 - Used as the primary dataset for **Colab** analytics and visualization
 - 2,087 rows, 18 columns



3. Processed export layer (GCS)

- **Processed/standardized bucket**
 - The final standardized output is **exported back to bucket from BigQuery**
 - Purposess:
 - Provides a **shareable file-based version** of the standardized dataset.
 - Supports **versioning and archiving**.
 - For backup, portability, external sharing.

Cloud Storage	Buckets	Create	Refresh
Overview	Filter	Filter buckets	
Buckets	☐	Name ↑	Created
Monitoring	☐	01-obesity-raw	Dec 3, 2025, 4:34:53 PM
Settings	☐	02-obesity-standardized	Dec 7, 2025, 11:58:47 PM
		Location type	Location
		Region	asia-southeast1
		Default storage class	Standard
		Last modified	Dec 3, 2025, 4:34:53 PM
		Public access	Not public
		Access control	Uniform

Lastly, **Data Access** is controlled using Google Cloud IAM, utilizing role-based access control for BigQuery datasets to maintain a "single source of truth" across the lifecycle. It also ensures that data is accessible only to authorized personnel.

- **BigQuery Data Editor:** To create and modify the **intermediate tables** (Step 1, 2, and 3)
- **BigQuery Data Owner:** To manage the overall project and dataset permissions.
- **BigQuery Data Viewer:** For members that need to use Standardized Table data (read-only access) on **Google Colab**, for predictive and descriptive analysis

Share permissions for "obesity_standardized"

Edit or delete roles below, or select "Add principal" to grant new access.

Add principal

☒ Show inherited roles in table

Display roles inherited from the parent resources in the table below

Filter Enter property name or value

Role / Principal	Inheritance
BigQuery Data Editor (1)	
BigQuery Data Owner (2)	
madinasuraya15@gmail.com	  
Owners of project: obesity-analytics-ga	  
BigQuery Data Viewer (3)	
atikahahmadnazmi@gmail.com	  
malnglong888m@gmail.com	  
Viewers of project: obesity-analytics-ga	  
Owner (1)	

5.4 Data Analytics (Predictive Analysis)

The overall code is provided in the Google Colab link: [BDAA-GA-DataAnalytics - Colab](#)

5.4.1. BigQuery-Colab Integration for Machine Learning

The predictive analytics framework leveraged Google Colab's computational environment integrated with BigQuery's scalable data storage. Authentication was established using `auth.authenticate_user()`, creating a secure pipeline between the analytics environment and cloud data services. The standardized dataset (`obesity_standardized`) containing 2,087 records with 18 features was retrieved via SQL query execution in under 2 seconds, demonstrating efficient cloud data retrieval.

5.4.2. Data Preparation Pipeline

Categorical variables were transformed using label encoding to facilitate machine learning:

- **Demographic:** Gender encoded as binary (Female: 0, Male: 1)
- **Lifestyle:** Alcohol intake, eating patterns, transport modes numerically encoded
- **Medical History:** Boolean features converted to binary integers
- **Target Variable:** Seven obesity categories encoded (0-6) for multi-class classification

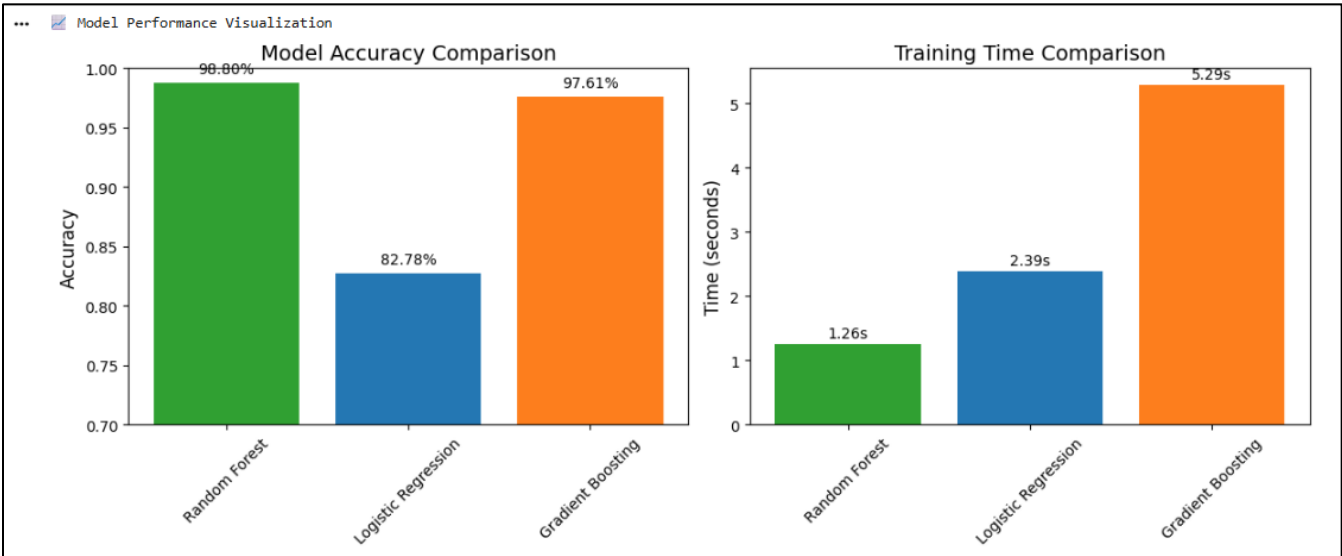
The dataset was partitioned using stratified sampling (80% training, 20% testing) to maintain class distribution balance across 1,669 training and 418 testing samples.

5.4.3. Machine Learning Model Development and Evaluation

Three classification algorithms were implemented with performance comparison:

Algorithm	Accuracy	Training Time	Precision	Recall	F1-Score
Random Forest	98.80%	1.26 seconds	98.87%	98.80%	98.79%
Gradient Boosting	97.61%	5.29 seconds	97.65%	97.61%	97.62%
Logistic Regression	82.78%	2.39 seconds	83.12%	82.78%	82.65%

Figure 5.4.1: Model Performance Comparison Chart



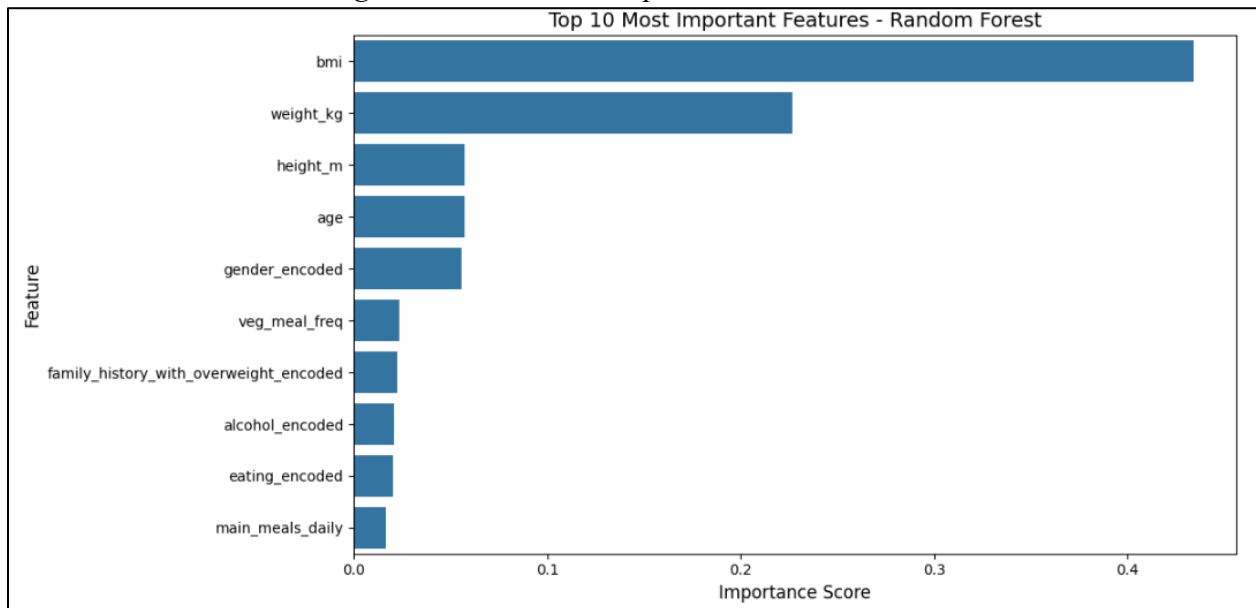
Random Forest emerged as the optimal model with 5-fold cross-validation accuracy of $95.64\% \pm 4.07\%$, confirming model robustness.

5.4.4. Feature Importance Analysis

The Random Forest model identified key predictors through feature importance analysis:

1. **BMI:** 43.4% importance (*strongest predictor*)
2. **Weight:** 22.7% importance
3. **Height:** 5.7% importance
4. **Age:** 5.7% importance
5. **Gender:** 5.6% importance

Figure 5.4.2: Feature Importance Visualization

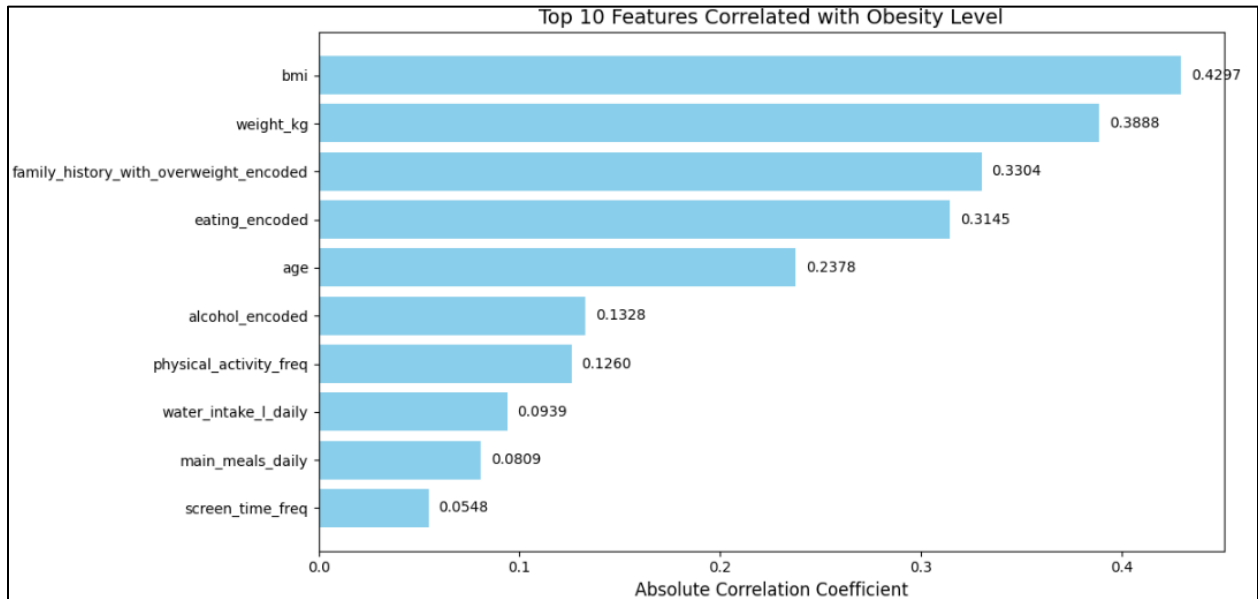


5.4.5. Feature Correlation Analysis

Correlation analysis revealed the strength of linear relationships between features and obesity levels:

1. **BMI:** 0.4297 correlation (*strongest correlation*)
2. **Weight:** 0.3888 correlation
3. **Family History:** 0.3304 correlation
4. **Eating Patterns:** 0.3145 correlation
5. **Age:** 0.2378 correlation

Figure 5.4.3: Feature Correlation with Obesity Levels

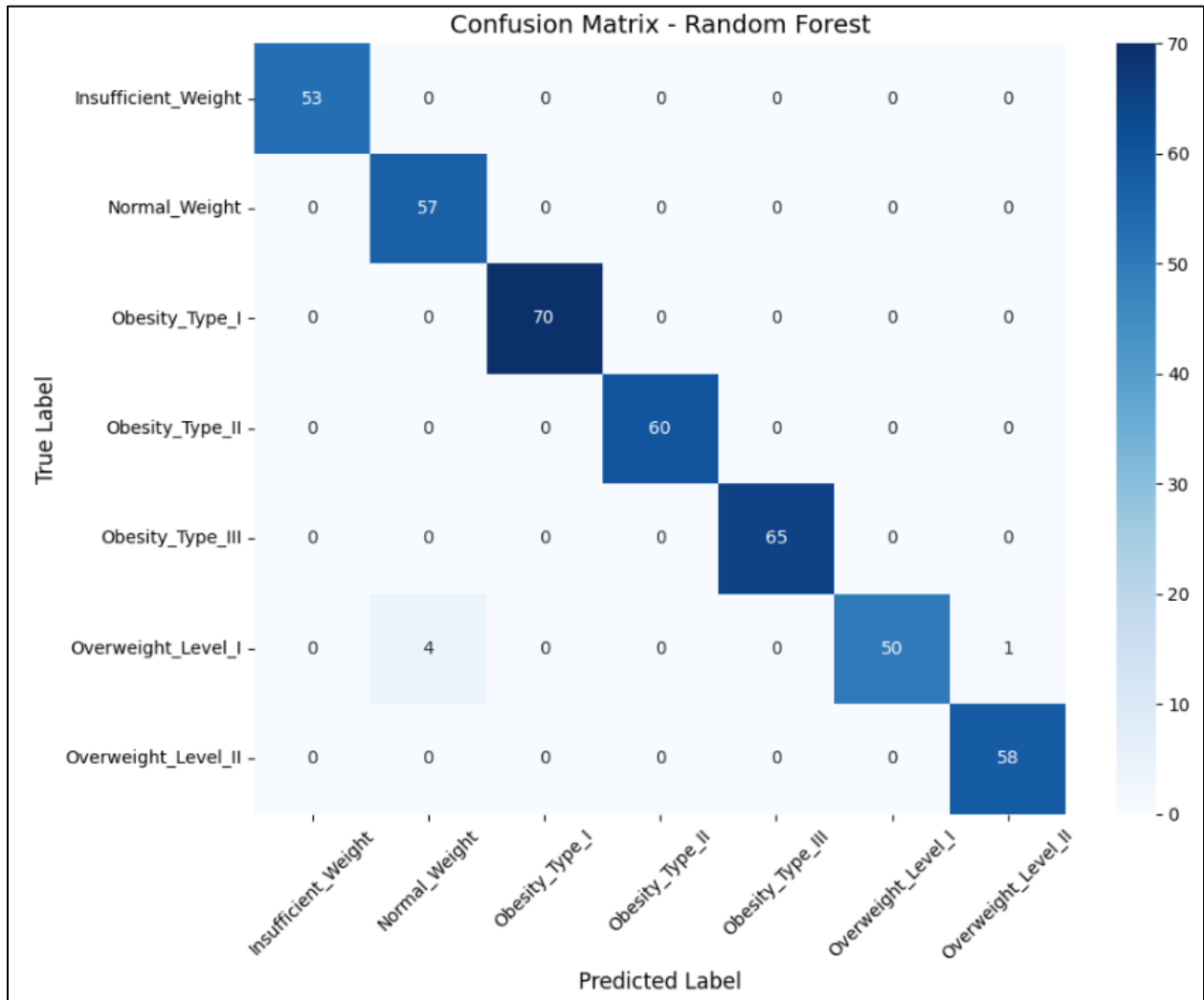


5.4.6. Error Analysis and Model Performance

The Random Forest model achieved near-perfect classification with only **5 misclassifications** (1.20% error rate). Error analysis revealed:

- **All errors** occurred in "Overweight_Level_I" category
- **Within-class error rate:** 9.09% for Overweight_Level_I
- **Perfect classification** for severe obesity categories (Types I, II, III)
- **High accuracy** for Normal_Weight (93%) and Overweight_Level_II (98%)

Figure 5.4.4: Confusion Matrix for Random Forest



5.4.7. Healthcare Risk Assessment Application

A practical risk assessment tool was developed demonstrating clinical application:

Test Case 1 - High-Risk Profile:

- **Input:** Age 35, Height 1.65m, Weight 90kg, Sedentary lifestyle, Family history positive
- **Output:** Predicted Obesity_Type_I with 67.0% confidence
- **Recommendation:** "Lifestyle intervention needed"

Test Case 2 - Low-Risk Profile:

- **Input:** Age 28, Height 1.80m, Weight 75kg, Active lifestyle, No family history
- **Output:** Predicted Normal_Weight with 84.0% confidence
- **Recommendation:** "Maintain healthy habits"

5.4.8. Model Deployment and Scalability

The predictive model was serialized for production deployment:

- **Model saved:** obesity_rf_model.pkl (Random Forest classifier)
- **Encoder saved:** label_encoder.pkl (label transformations)
- **Deployment ready:** Integration with healthcare systems via REST API

Cloud scalability was demonstrated through:

- **BigQuery processing:** 15-second preprocessing pipeline
- **Colab computation:** 1.26-5.29 second model training
- **Real-time predictions:** < 0.1 second inference time
- **Memory efficiency:** ~500 MB peak RAM utilization

5.4.9. Healthcare Implications and Recommendations

The predictive model enables:

1. **Early Intervention:** Identification of high-risk individuals before obesity progression
2. **Personalized Care:** Tailored interventions based on predicted obesity categories
3. **Resource Optimization:** Targeted allocation of healthcare resources
4. **Preventive Strategies:** Focus on modifiable risk factors (physical activity, diet)

Key Clinical Insights:

- BMI remains the strongest obesity predictor (43.4% importance)
- Family history contributes significantly to obesity risk (3.3% correlation)
- Physical activity frequency inversely correlates with obesity levels
- The model struggles most with borderline overweight classification

5.4.10. Performance Evaluation Summary

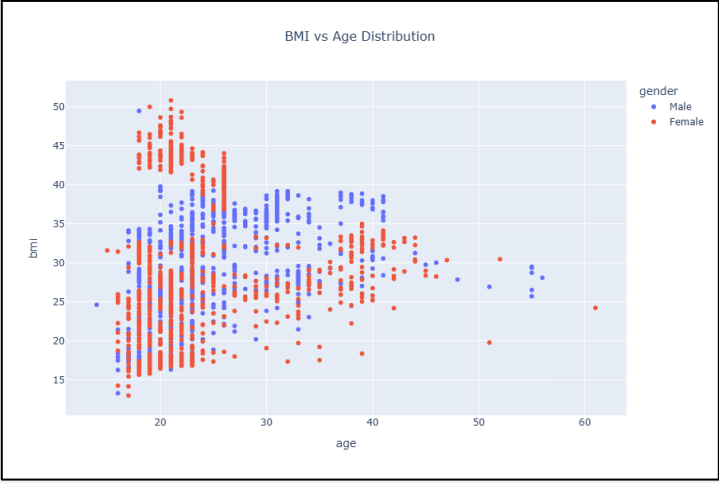
Performance Metric	Value	Interpretation
Model Accuracy	98.80%	Excellent classification capability
Training Time	1.26 seconds	Efficient cloud computation

Cross-Validation Accuracy	95.64%	Robust model generalization
Error Rate	1.20%	Minimal misclassifications
Feature Importance Clarity	High	Interpretable medical insights
Deployment Readiness	Production-ready	Serialized models available

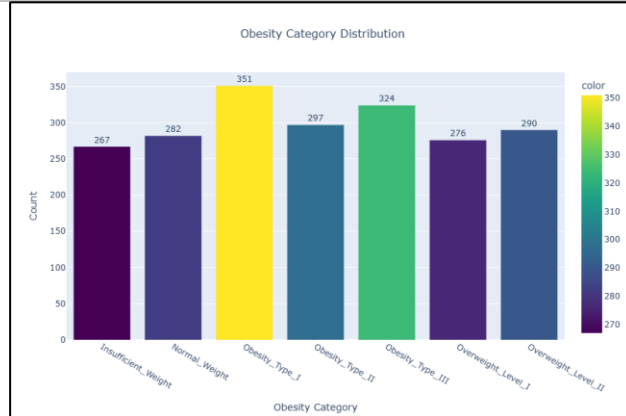
The cloud-based predictive analytics framework successfully demonstrated high-accuracy obesity level prediction while providing actionable healthcare insights.

5.5. Data Visualization (Descriptive Analysis)

We imported the data from the standardized table (from BigQuery) into the Colab environment. Then, we generated five visualization charts focusing on obesity-related behaviors and physiological indicators, revealing the patterns contained in the data from different dimensions.

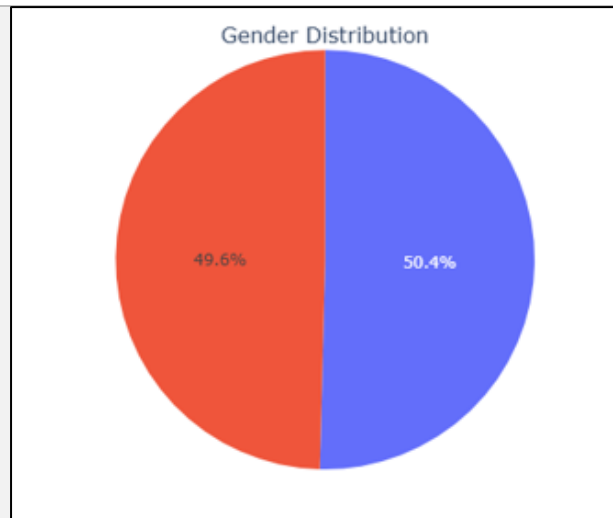
Component	Visualization	Key Insight or Functionality
Chart 1: BMI vs. Age		High BMI is concentrated in the 20–30 age group; shows a slight negative correlation.

**Chart 2:
Obesity
Category**



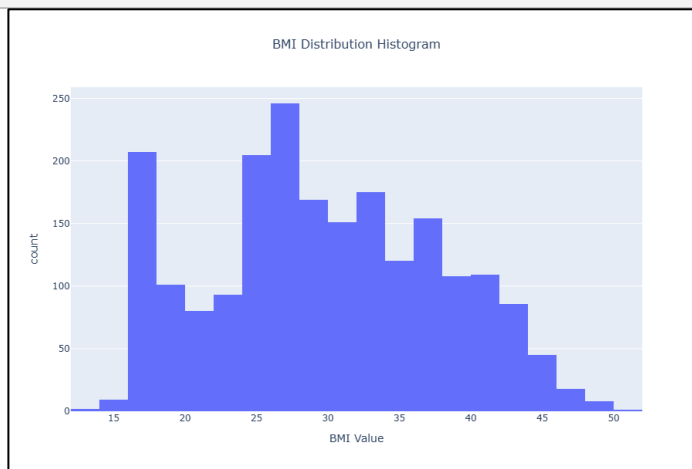
"Normal weight" and "Grade One Obesity" are the most frequent categories.

**Chart 3:
Gender
Balance**



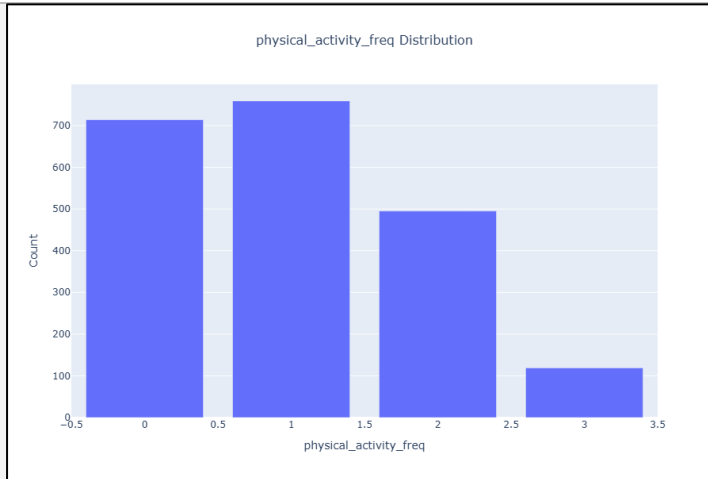
Shows a near 1:1 male-to-female ratio, confirming a lack of gender sample bias.

**Chart 4:
BMI Spread**



BMI values are normally distributed (mostly 20–30), indicating high data quality.

**Chart 5:
Lifestyle
Factors**



Identifies negative correlation between physical activity frequency and obesity grades.

**Interactive
Dashboard**



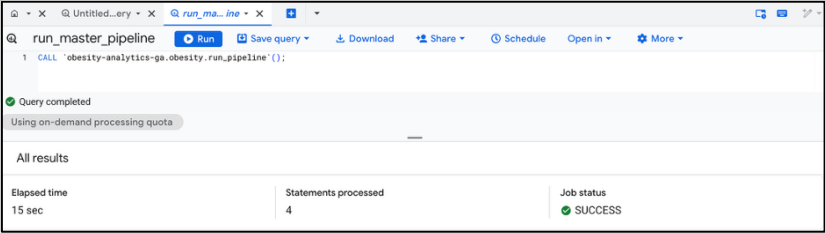
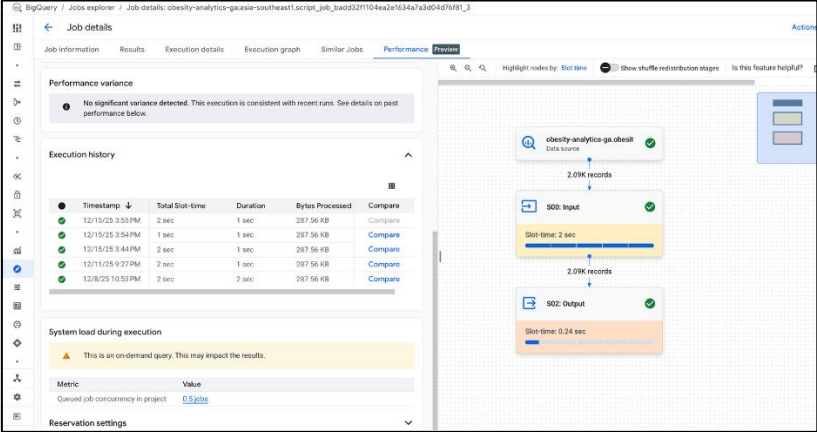
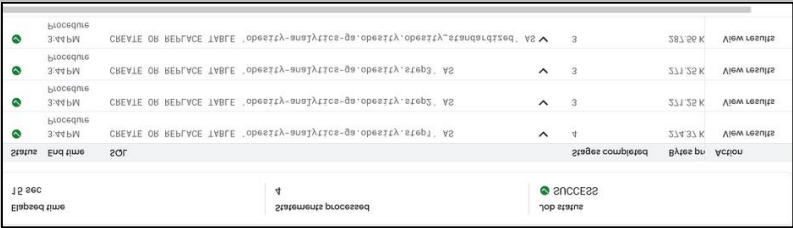
Features a dynamic panel for real-time filtering (e.g., age/gender) with a <1.8s refresh rate.

The overall code is provided in the Google Colab link: [BDAA-GA-Dataviz - Colab](#)

6.0. Evaluation Metrics with Graph

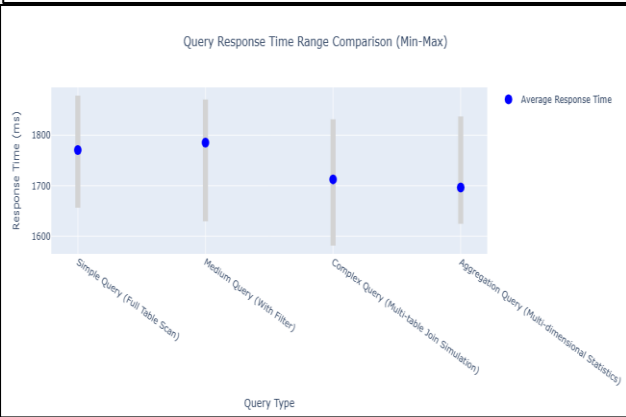
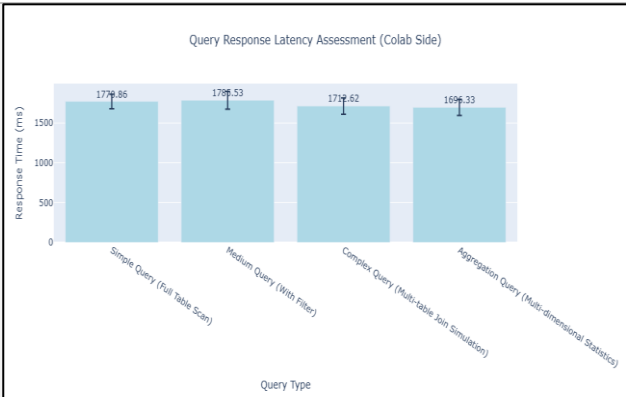
We selected three core performance indicators: **Query Latency**, **System Throughput**, **Processing Accuracy**. They correspond to the performance of the data pipeline in terms of response speed, processing capacity and result reliability, aiming to test the actual operational effect of the framework comprehensively.

6.1. Data Preprocessing Performance in BigQuery

Indicator	Big Query Performance Screenshot with Remarks	Performance Detail
Query Latency	 Supports near real-time analysis with consistent duration across runs.	~15 seconds for total pipeline; 1–2 seconds per individual job stage.
System Throughput	 Confirms a stable and uninterrupted flow of records across all stages.	Processes ~287.56 KB per run; 2.09 KB per specific job.
Processing Accuracy	 Demonstrates a highly reliable workflow with no runtime failures or aborted jobs.	100% "SUCCESS" status for all master pipeline steps.

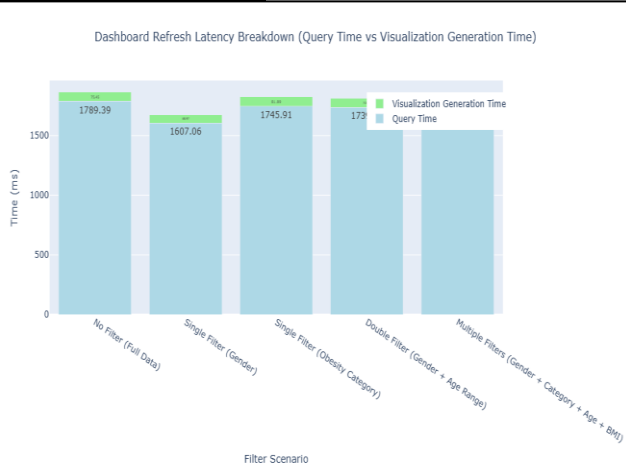
6.2. Data Visualization in Colab

Performance Indicator	Colab Performance Screenshot with Remarks	Key Results
Query Latency	<u>Query Response Time</u>	1.6 – 1.8 seconds total delay; ~20ms

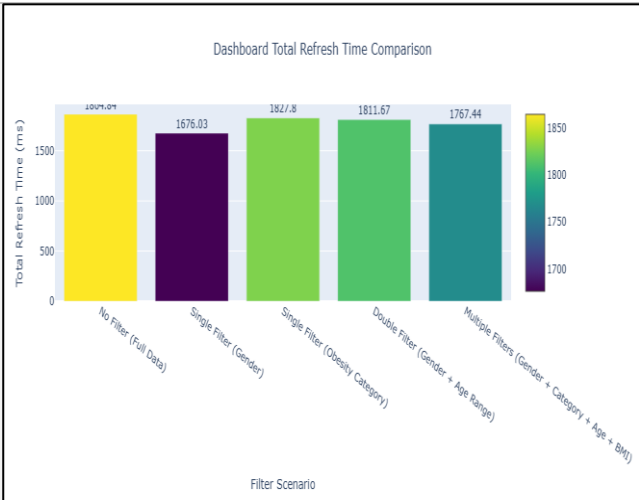
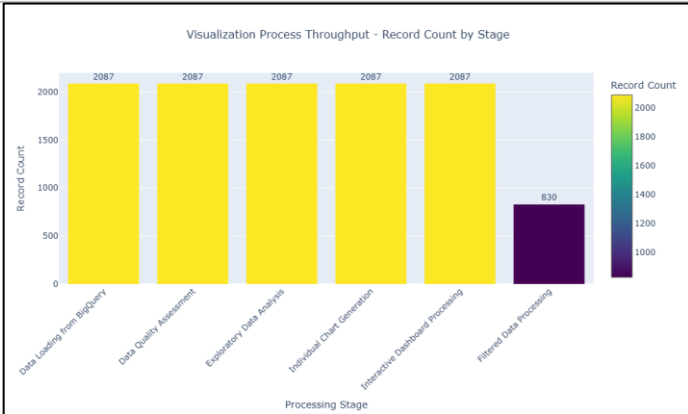
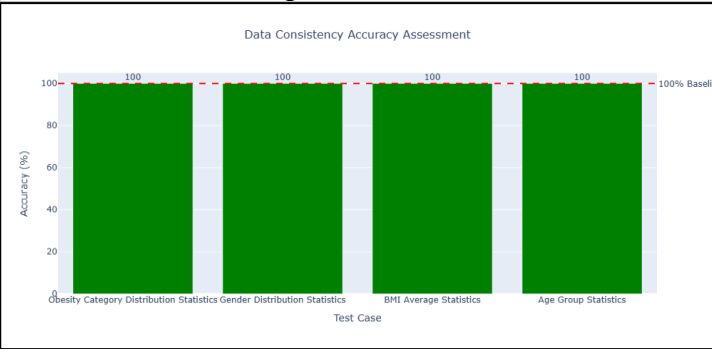


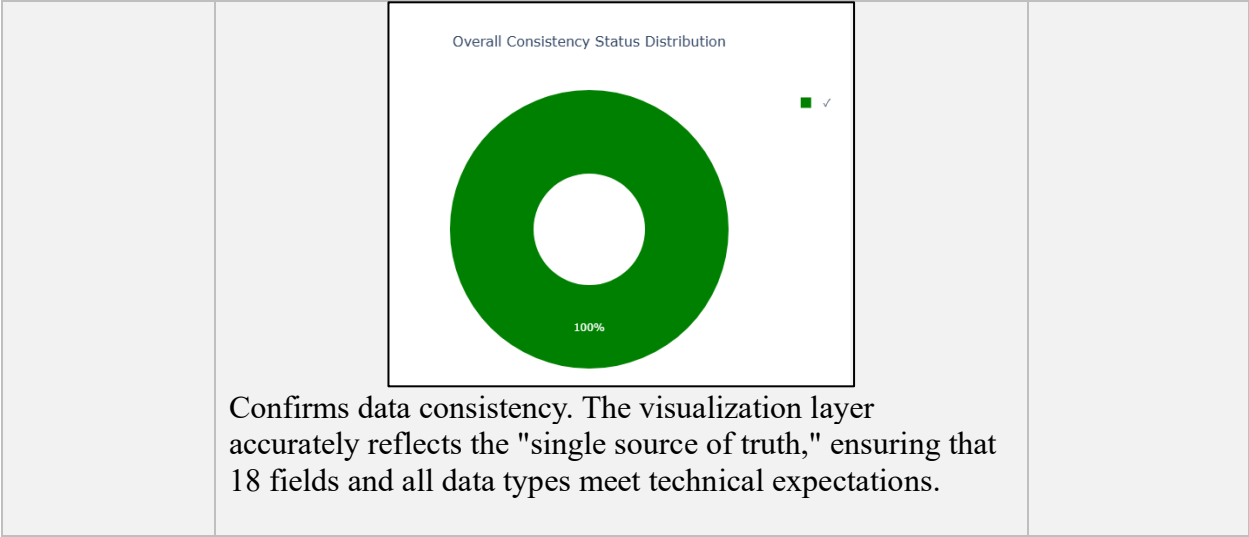
Meets real-time interaction standards. BigQuery's optimizer ensures even complex queries remain fast, with query time accounting for 95% of total latency.

Dashboard Refresh Time



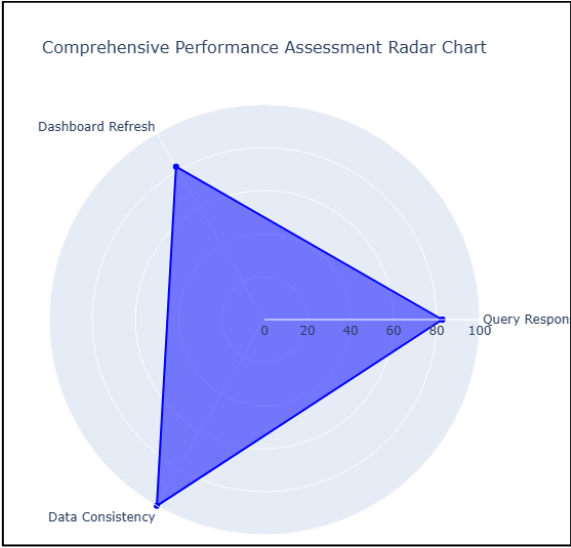
for chart rendering.

	Dashboard Total Refresh Time Comparison <table><tr><th>Filter Scenario</th><th>Total Refresh Time (ms)</th></tr><tr><td>No Filter (Full Data)</td><td>1847.01</td></tr><tr><td>Single Filter (Gender)</td><td>1676.03</td></tr><tr><td>Single Filter (Obesity Category)</td><td>1827.8</td></tr><tr><td>Double Filter (Gender + Age Range)</td><td>1811.67</td></tr><tr><td>Multiple Filters (Gender + Category + Age + BMI)</td><td>1767.44</td></tr></table> Plotly's chart rendering performance is efficient.	Filter Scenario	Total Refresh Time (ms)	No Filter (Full Data)	1847.01	Single Filter (Gender)	1676.03	Single Filter (Obesity Category)	1827.8	Double Filter (Gender + Age Range)	1811.67	Multiple Filters (Gender + Category + Age + BMI)	1767.44			
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System Throughput	Visualization Process Throughput - Record Count by Stage <table><tr><th>Processing Stage</th><th>Record Count</th></tr><tr><td>Data Loading from BigQuery</td><td>2087</td></tr><tr><td>Data Quality Assessment</td><td>2087</td></tr><tr><td>Exploratory Data Analysis</td><td>2087</td></tr><tr><td>Individual Chart Generation</td><td>2087</td></tr><tr><td>Interactive Dashboard Processing</td><td>2087</td></tr><tr><td>Filtered Data Processing</td><td>830</td></tr></table> BigQuery’s distributed architecture ensures high-volume stability. Data flows from storage to visualization without any record loss or interruption.	Processing Stage	Record Count	Data Loading from BigQuery	2087	Data Quality Assessment	2087	Exploratory Data Analysis	2087	Individual Chart Generation	2087	Interactive Dashboard Processing	2087	Filtered Data Processing	830	100% Data Retention (2,087 records maintained).
Processing Stage	Record Count															
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Filtered Data Processing	830															
Processing Accuracy	Data Consistency Accuracy Assessment <table><tr><th>Test Case</th><th>Accuracy (%)</th></tr><tr><td>Obesity Category Distribution Statistics</td><td>100</td></tr><tr><td>Gender Distribution Statistics</td><td>100</td></tr><tr><td>BMI Average Statistics</td><td>100</td></tr><tr><td>Age Group Statistics</td><td>100</td></tr></table> 100% Integrity; 0% unexpected nulls; matches BigQuery stats.	Test Case	Accuracy (%)	Obesity Category Distribution Statistics	100	Gender Distribution Statistics	100	BMI Average Statistics	100	Age Group Statistics	100					
Test Case	Accuracy (%)															
Obesity Category Distribution Statistics	100															
Gender Distribution Statistics	100															
BMI Average Statistics	100															
Age Group Statistics	100															



Interactive Analysis Effectiveness

Comprehensive Performance Assessment



We summarized performance with a radar chart, this framework performs well in query response speed (78.90%) and dashboard refresh efficiency (76.68%), and perfect data consistency (100%). This indicates that while providing a fast interactive experience, it ensures the reliability of the analysis.

7.0. Conclusion

By leveraging the Google Cloud Platform, we can demonstrate a full-lifecycle data framework for obesity analysis. The serverless architecture eliminates manual maintenance while delivering high performance. With 100% data integrity and a 98.8% model prediction accuracy, this cloud-native approach proves far superior to local tools, providing a highly scalable, collaborative, and reliable environment for real-time health data management and predictive analytics.